

சொல் AI (SOL AI)

Tamil Lexical Intelligence & Classical Literary Grounding System

Official Browser Extension Manual • Manifest V3

Version: 1.0.0

Platform: Chrome / Edge / Brave

1. How to Download & Install the Extension

- Download Files:** Click `Download சொல் AI Extension` on the website to receive both the ZIP bundle and this PDF guide.
- Extract Archive:** Right-click `sol-ai-extension.zip` and select **Extract All** to an easily accessible folder on your computer.
- Open Extensions Page:** In Google Chrome, Microsoft Edge, or Brave, navigate to `chrome://extensions` (or `edge://extensions`).
- Enable Developer Mode:** Toggle the **Developer mode** switch at the top-right corner of the page to **ON**.
- Load Extension:** Click **"Load unpacked"** in the top-left corner and select the extracted extension folder. The சொல் AI emblem will appear in your toolbar!

2. How to Use on Any Webpage

Instant Highlight & Lookup

Highlight any Tamil word or compound phrase on any webpage (news portals, Wikipedia, blogs). Click the floating சொல் AI badge to open the inspection drawer.

FOMA FST Morphology

Deconstructs complex agglutinative words into their base lemma, grammatical case, number, and suffix attachments with zero guesswork.

Deterministic Contextual WSD

Disambiguates polysemous words (e.g., கால் as fraction vs. anatomical foot) based on the exact enclosing sentence context.

Classical Literary Grounding

Retrieves verified historical couplets and verses from Sangam literature (Tirukkural, Purananuru, Manimekalai, Silappatikaram) without hallucination.

3. System Limitations & Usage Policy

20 Requests Per Day Usage Policy

- Daily Rate Limit:** To ensure equitable access and protect shared cloud infrastructure, each user is allotted a policy limit of **20 requests per day**.
- AWS & Free-Tier AI APIs:** The backend is deployed on an AWS EC2 instance connected to free-tier LLM endpoints (Google Gemini & Groq).
- Responsible Use:** Please use lookups thoughtfully, allow 2–3 seconds between queries, and avoid automated scripts or rapid clicking.
- Deterministic Safe Fallback:** If third-party AI rate limits are exceeded, SOL AI automatically falls back to our offline dictionary & Sangam database engine.

4. Future Roadmap & Improvements



Chrome Web Store Release

Official store publication for one-click installation and automatic background updates.



Corpus & Resource Expansion

Integration of further Sangam, Pathinenkilkanakku, and contemporary Tamil lexicons.



Client-Side WebAssembly (WASM)

Compiling ThamizhiMorph FSTs to WASM for 100% offline, zero-latency in-browser parsing.



Dialectal & Idiom Intelligence

Advanced disambiguation for regional, Sri Lankan, and diaspora Tamil idioms.